

Intermediate Microeconomics — Week 2, Day 1

In-class Exercises: Supply, Demand, Equilibrium, Elasticity

Perloff, Ch. 2 • work in pairs • solutions at the end

Exercise 1 — Equilibrium and a supply shock [competency: solve & do comparative statics]

A regional market for a commodity has

$$Q_d = 200 - 5P, \quad Q_s = -40 + 3P,$$

with P in \$ and Q in thousand units.

- (a) Find the equilibrium price and quantity.
 - (b) A cheaper input shifts supply to $Q'_s = -16 + 3P$. Find the new equilibrium and report ΔP and ΔQ . In one sentence, is this the “demand-shock” or “supply-shock” signature?
 - (c) Compute the price elasticity of demand at the *original* equilibrium. Is demand elastic or inelastic there?
 - (d) Using (c), explain qualitatively why the price moved as it did in (b): would a *more* inelastic demand have made ΔP larger or smaller?
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Exercise 2 — Elasticity, revenue, and a constant-elasticity curve [competency: elasticity & revenue, not equilibrium-solving]

Part A — linear demand. A firm faces $Q = 180 - 3P$.

- (a) Compute the price elasticity of demand at $P = 20$ and at $P = 40$. Classify each.
- (b) Find the price that maximises revenue $R = PQ$, and the revenue there. Confirm it is the unit-elastic point.

Part B — constant-elasticity demand. Now suppose instead $Q = 1200 P^{-2}$.

- (c) Show that the price elasticity is the same at every price, and state its value.
 - (d) Compute Q and revenue R at $P = 10$ and $P = 20$. Does revenue rise or fall when price rises? Reconcile with your elasticity in (c).
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Reconvene: for each exercise be ready to state the *sign* and the *size* of the answer, and one sentence on what the elasticity told you.